Cross-Platform App Development

Exercise – Event Booking Mobile Application

Scenario

A company is planning to develop a mobile application called **EventHub** to help users discover and book events and activities available in their area.

The application should allow users to browse different events, view event information, make bookings, and manage their bookings. Event organizers should also be able to publish and manage their events.

You are required to design and develop a **cross-platform mobile application** based on this scenario.

Development Requirements

Develop the application using **ONE** of the following frameworks:

- **Flutter**
- **React Native**

You may use **Dart** with Flutter or **JavaScript/TypeScript** with React Native.

You are required to follow the **official documentation** of the selected framework when developing the application.

Flutter: Official Flutter Documentation

React Native: Official React Native Documentation

Application Requirements

Your application should provide the following functionality.

1. User Account

Users should be able to:

- Create an account
- Log in
- Log out
- View and update their profile

The application should validate user input appropriately.

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2. Browse Events

Users should be able to view available events.

Each event should provide relevant information such as:

- Event name
- Event image
- Description
- Date and time
- Location
- Category
- Price
- Available seats

Users should be able to **search for events** and **filter events by category**.

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3. Event Details

When a user selects an event, the application should display a detailed view of the event.

The user should be able to:

- View complete event information
- View the event location
- Check availability
- Proceed with a booking

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4. Event Booking

Users should be able to book an event through the application.

The booking process should:

- Collect the necessary information from the user
- Validate the information
- Confirm the booking
- Store the booking information
- Display the booking confirmation

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5. My Bookings

Users should be able to view their current and previous bookings.

The application should display appropriate booking information and status.

Users should also be able to cancel a booking where applicable.

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6. Event Organizer

An organizer should be able to manage events.

The application should allow an organizer to:

- Add a new event
- View their events
- Edit an event
- Remove an event
- View bookings for an event

Advanced UI Requirements

The application should demonstrate appropriate **mobile UI design and navigation**.

You should use suitable:

- Screen navigation
- Navigation bars/tabs/drawers
- List and grid views
- Card-based layouts
- Scrollable views
- Forms
- Dialogs
- Notifications
- Responsive layouts

The interface should be **consistent, user-friendly, and suitable for a mobile device**.

You are expected to make your own decisions regarding the appropriate UI components and navigation structure.

Database and API Requirements

The application must use a **database** to store information required by the application.

At minimum, the application should store information related to:

- Users
- Events
- Bookings

The application must also communicate with a **REST API** to retrieve and/or update application data.

For example, the application may use APIs to:

- Retrieve available events
- Retrieve event details
- Create bookings
- Update bookings
- Manage events

The application should handle the API responses and display the relevant information to the user.

Notifications

The application should provide appropriate notifications to users.

For example:

- Booking confirmation
- Booking cancellation
- Event reminder
- Event update

Notifications should be used where they are meaningful to the application.

Local Data

The application should store at least some relevant information locally on the device.

For example, you may use local storage for:

- User session information
- Favourite events

- User preferences

Choose a suitable use of local storage based on your application design.

Application Quality

Your application should:

- Have a clear and consistent user interface
- Provide smooth navigation between screens
- Validate user input
- Display appropriate feedback after user actions
- Handle API/database errors appropriately
- Avoid application crashes during normal use
- Work correctly on a mobile device/emulator

Student Freedom

You are **not required to follow a predefined screen design**.

You should make your own decisions regarding:

- Application name and branding
- UI design
- Colour scheme
- Screen layout
- Navigation structure
- Database technology
- API implementation
- State management approach
- Suitable libraries/packages

Your implementation decisions should be appropriate for the given scenario.

Submission

- Create a **GitHub repository** for your project.
- Push the complete project source code to the GitHub repository.
- Copy the **GitHub repository link** and paste it into the **designated submission location** provided for this exercise.